

On Generalized Wishart Distributions - II: Sphericity Test

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Abstract

In this second part of the paper, we make use of the distribution of the trace of a generalized Wishart matrix based on elliptical models to derive the moments of statistic V used for testing sphericity for a general elliptical model. From the general expressions, we derive specific expressions for the special case of the Kotz family, which includes the Gaussian subfamily. Finally, to illustrate the usefulness of the approach, the exact distribution of the statistic V is derived in terms of the G -function by using Mellin transform and complex integration techniques.

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1 Introduction

In the first part of this paper, we defined and derived some properties of the generalized Wishart distribution, based on an elliptical model. We then used it to study the distribution of the likelihood ratio statistic for testing the homogeneity of covariance matrices. In this second part of the paper, we study the sphericity test under elliptical models and derive the exact distribution and moments of the test statistic for general elliptical models. Specifically, we consider the problem of testing the hypothesis

$$H_0 : \Sigma_1 = \lambda \mathbf{I}_m$$

under the assumption that the sample comes from an elliptical model.

The sphericity test has been studied extensively by many authors under the classical Gaussian model. Muirhead (1982) reviewed all the developments on the sphericity test under multivariate normality and also presented some asymptotic results. His unified treatment includes expressions in terms of series of zonal polynomials, which at that time were difficult to compute and so necessitated the asymptotic approach. After the work of Koev and Edelman (2006), the computation of the exact densities of test statistics derived under multivariate normality can be carried out.

However, with regard to extensions to elliptical models, very limited literature exists discussing primarily some general properties of invariance, unbiasedness, and so on [see, for example, Fang and Anderson (1990), Fang and Zhang (1990) and Gupta and Varga (1993)], but no results are available on explicit expressions for the distributions of the test statistics in the general case.

In Part I of this paper, we presented the generalized Wishart distribution and discussed some of its properties and its application to testing the homogeneity of covariance matrices. Here, we use it to study the sphericity test for a general elliptical model. In Section 2, we use the generalized Wishart distribution to study the distribution of the trace of a Wishart matrix under any elliptical model. The classical Gaussian result is derived as a simple Corollary and the results for a Kotz non-Gaussian model are also obtained as a special case. In Section 3, the moments of the sphericity test statistic V are derived. The corresponding Gaussian results are obtained also as a Corollary. Here again, results for the Kotz distribution are developed as a special case. These results are based on a new integral representation of generalized zonal polynomial series. Finally, the exact non-null distribution of the V statistic under a Gaussian model is derived in Section 4. The required densities are obtained by the use of Mellin transform, and are expressed in terms of G-functions.

2 Distribution of the Trace of a Generalized Wishart Matrix

Recall that an $n \times m$ matrix $\mathbf{X}$ is said to have a matrix-variate elliptically contoured distribution, denoted by $E_{n \times m}(\boldsymbol{\mu}, \boldsymbol{\Theta} \otimes \boldsymbol{\Sigma}, h)$, if its density function is given by

$$g_{\mathbf{X}}(\mathbf{X}) = \frac{1}{|\boldsymbol{\Sigma}|^{n/2} |\boldsymbol{\Theta}|^{m/2}} h \left(\text{tr } \boldsymbol{\Theta}^{-1} (\mathbf{X} - \boldsymbol{\mu}) \boldsymbol{\Sigma}^{-1} (\mathbf{X} - \boldsymbol{\mu})' \right),$$

where $\boldsymbol{\Sigma} : m \times m$, $\boldsymbol{\Theta} : n \times n$, $\boldsymbol{\Sigma} > \mathbf{0}$, $\boldsymbol{\Theta} > \mathbf{0}$, and the function $h : \Re \rightarrow [0, \infty)$, termed the generator function, is such that $\int_0^\infty u^{nm-1} h(u^2) du < \infty$.

Let us denote the Elliptical Wishart $EW_m(n, \Sigma, h)$ as the distribution of the $m \times m$ random matrix $\mathbf{A} = \mathbf{Z}'\mathbf{Z}$, where the $n \times m$ matrix $\mathbf{Z}$ is $E_{n \times m}(\mathbf{0}, \mathbf{I}_n \otimes \Sigma, h)$ and $n \geq m$; then from Part I of this paper, the density function of $\mathbf{A}$ is

$$\frac{\pi^{mn/2}}{\Gamma_m\left(\frac{1}{2}n\right) |\Sigma|^{n/2}} |\mathbf{A}|^{(n-m-1)/2} h(\text{tr } \Sigma^{-1} \mathbf{A}),$$

where $\Gamma_m(a) = \pi^{m(m-1)/4} \prod_{i=1}^m \Gamma[a - 1/2(i-1)]$ and $\Re(a) > 1/2(m-1)$.

Let the random matrix $\mathbf{A}$ be distributed as $EW_m(n, \Sigma, h)$. Then, the statistic

$$V = \frac{|\mathbf{A}|}{\left(\frac{1}{m} \text{tr } \mathbf{A}\right)^m}$$

is commonly referred to as the ellipticity test statistic in the multivariate normal case.

Information about the distribution of the ellipticity test statistic V can be obtained through its moments. For this purpose, we need the distribution of $\text{tr } \mathbf{A}$, when $\mathbf{A}$ is $EW_m(n, \Sigma, h)$. In the multivariate normal case when $m = 2$, this is known to be a mixture of gamma distributions. In general, it can be expressed as an infinite sum of zonal polynomials.

Now, we can present the main result of this section.

THEOREM 2.1. *If $\mathbf{A}$ is $EW_m(n, \Sigma, h)$ with $n \geq m$, the distribution function of $\text{tr } \mathbf{A}$ can be expressed as*

$$P(\text{tr } \mathbf{A} \leq x) = \pi^{mn/2} \sum_{k=0}^{\infty} \frac{1}{k!} \int_{-\infty}^x r(u, k) du,$$

where the density function $r(u, k)$ is obtained through an inversion of the moment generating function

$$M_u(t, k, h) = \frac{\int_0^{\infty} h(w) w^{\frac{mn}{2} + k - 1} dw}{\Gamma\left(\frac{mn}{2} + k\right)} \sum_{\kappa} \left(\frac{n}{2}\right)_{\kappa} C_{\kappa}(t\Sigma),$$

with $C_{\kappa}(\mathbf{B})$ being the zonal polynomials of the positive definite $\mathbf{B}$ associated with the ordered partition $\kappa = (k_1, k_2, \dots)$ of k (see Muirhead (1982)) and $(a)_{\kappa} = \pi^{m(m-1)/4} \prod_{i=1}^m \Gamma[a + k_i - 1/2(i-1)] / \Gamma_m(a)$, $\Re(a) > 1/2(m-1)$.

PROOF. Consider the moment generating function

$$\phi(t) = E(\text{tr}(t\mathbf{A})) = \frac{\pi^{mn/2}}{\Gamma_m\left(\frac{n}{2}\right) |\Sigma|^{n/2}} \int_{\mathbf{A} > \mathbf{0}} h(\text{tr } \Sigma^{-1} \mathbf{A}) |\mathbf{A}|^{(n-m-1)/2} \text{etr}(t\mathbf{A}) (d\mathbf{A})$$

associated with the elliptical density of $\mathbf{A}$ presented in Part I of the paper. Upon expanding $\text{etr}(t\mathbf{A})$ in terms of zonal polynomials and then using the integral result of Caro-Lopera, Díaz-García and González-Farías et al. (2009, Theorem 1), we obtain

$$\begin{aligned}\phi(t) &= \frac{\pi^{mn/2}}{\Gamma_m\left(\frac{n}{2}\right) |\Sigma|^{n/2}} \sum_{k=0}^{\infty} \frac{1}{k!} \sum_{\kappa} \int_{\mathbf{A} > \mathbf{0}} h(\text{tr } \Sigma^{-1} \mathbf{A}) |\mathbf{A}|^{(n-m-1)/2} C_{\kappa}(t\mathbf{A}) (d\mathbf{A}) \\ &= \pi^{mn/2} \sum_{k=0}^{\infty} \frac{\int_0^{\infty} h(w) w^{\frac{mn}{2}+k-1} dw}{k! \Gamma\left(\frac{mn}{2} + k\right)} \sum_{\kappa} \left(\frac{n}{2}\right)_{\kappa} C_{\kappa}(t\Sigma),\end{aligned}$$

which is the required result.

The corresponding result for the Gaussian case [see Muirhead (1982) and Khatri and Srivastava (1971)] is now deduced from Theorem 2.1.

COROLLARY 2.1. *If $\mathbf{A}$ is $W_m(n, \Sigma)$ with $n \geq m$, the distribution function of $\text{tr } \mathbf{A}$ can be expressed in the form*

$$P(\text{tr } \mathbf{A} \leq x) = |\lambda^{-1} \Sigma|^{-n/2} \sum_{k=0}^{\infty} \frac{1}{k!} P\left(\chi_{mn+2k}^2 \leq \frac{x}{\lambda}\right) \sum_{\kappa} \left(\frac{n}{2}\right)_{\kappa} C_{\kappa}(\mathbf{I} - \lambda \Sigma^{-1}), \quad (2.1)$$

where $0 < \lambda < \infty$ is arbitrary.

PROOF. In the Gaussian case, we have

$$h(y) = \frac{1}{(2\pi)^{mn/2}} e^{-y/2}$$

and

$$\int_0^{\infty} h(w) w^{\frac{mn}{2}+k-1} dw = 2^k \pi^{-\frac{mn}{2}} \Gamma\left(\frac{mn}{2} + k\right),$$

and consequently

$$\phi(t) = \sum_{k=0}^{\infty} \frac{1}{k!} \sum_{\kappa} \left(\frac{n}{2}\right)_{\kappa} C_{\kappa}(2t\Sigma) = |\mathbf{I} - 2t\Sigma|^{-\frac{n}{2}}.$$

Using now the fact that

$$\mathbf{I} - 2t\Sigma = (1 - 2t\lambda) \lambda^{-1} \Sigma \left(\mathbf{I} - \frac{1}{1 - 2t\lambda} (\mathbf{I} - \lambda \Sigma^{-1}) \right)$$

for arbitrary $0 < \lambda < \infty$, the moment generating function can be inverted and the required result in (2.1) then follows.

For explicit non-Gaussian models, the corresponding result is not available in the literature, but Theorem 2.1 would enable us to develop this result. For example, we now derive the distribution of the trace in the case of the Kotz model for which

$$h(y) = \frac{R^{T-1+\frac{mn}{2}} \Gamma\left(\frac{mn}{2}\right)}{\pi^{\frac{mn}{2}} \Gamma\left(T-1+\frac{mn}{2}\right)} y^{T-1} e^{-Ry},$$

In the special case of $T = 2$ and $R > 0$, we have

$$h(y) = 2(mn)^{-1} \pi^{-\frac{mn}{2}} R^{1+\frac{mn}{2}} y e^{-Ry}, \quad y > 0,$$

and

$$\frac{\int_0^\infty h(w) w^{\frac{mn}{2}+k-1} dw}{\Gamma\left(\frac{mn}{2} + k\right)} = 2(mn)^{-1} \pi^{-\frac{mn}{2}} R^{-k} \left(k + \frac{mn}{2}\right).$$

Consequently,

$$\begin{aligned} \phi(t) &= \sum_{k=0}^{\infty} \frac{1}{k!} \sum_{\kappa} \left(\frac{n}{2}\right)_{\kappa} C_{\kappa} \left(\frac{t}{R} \Sigma\right) \\ &\quad + 2(mn)^{-1} \sum_{k=0}^{\infty} \frac{k}{k!} \sum_{\kappa} \left(\frac{n}{2}\right)_{\kappa} C_{\kappa} \left(\frac{t}{R} \Sigma\right). \end{aligned} \quad (2.2)$$

It is of interest to observe that the moment generating function in (2.2) consists of two terms, with the first part corresponding to the Kotz type I model (including the Gaussian case when $R = 1/2$) and the second part that is worth studying further.

Recall that

$$\text{tr}(\mathbf{Y}) \text{etr } \mathbf{Y} = \sum_{k=0}^{\infty} \frac{k}{k!} \sum_{\kappa} C_{\kappa}(\mathbf{Y}),$$

and so it is easy to see that an integral representation for the second term (except for the constant) in (2.2) is given by

$$\begin{aligned} S &= \sum_{k=0}^{\infty} \frac{k}{k!} \sum_{\kappa} \left(\frac{n}{2}\right)_{\kappa} C_{\kappa} \left(\frac{t}{R} \Sigma\right) \\ &= \frac{|R \Sigma^{-1}|^a}{\Gamma_m(a)} \int_{\mathbf{A} > \mathbf{0}} \text{etr}(-\mathbf{A} R \Sigma^{-1}) |\mathbf{A}|^{a-(m+1)/2} \text{etr}(t \mathbf{A}) \text{tr}(t \mathbf{A}) (d\mathbf{A}) \end{aligned}$$

with $a = n/2$. For an arbitrary $0 < \lambda < \infty$, we have

$$\begin{aligned} S &= \frac{|R\Sigma^{-1}|^a}{\Gamma_m(a)} \int_{\mathbf{A} > \mathbf{0}} \text{etr} \left(-R\mathbf{A}\Sigma^{-1} \left(\mathbf{I} - \frac{t}{R}\Sigma \right) \right) |\mathbf{A}|^{a-(m+1)/2} \text{tr}(t\mathbf{A})(d\mathbf{A}) \\ &= \frac{|R\Sigma^{-1}|^a t}{\Gamma_m(a)} \int_{\mathbf{A} > \mathbf{0}} |\mathbf{A}|^{a-(m+1)/2} \text{etr} \left(-R\mathbf{A} \left(1 - \frac{t}{R}\lambda \right) \lambda^{-1} \right) \text{tr}(\mathbf{A}) \\ &\quad \times \text{etr} \left(R\mathbf{A}\lambda^{-1} (\mathbf{I} - \lambda\Sigma^{-1}) \right) (d\mathbf{A}). \end{aligned}$$

Expanding now in terms of zonal polynomials, we obtain

$$\begin{aligned} S &= \frac{|R\Sigma^{-1}|^a t \lambda}{\Gamma_m(a) R \left(1 - \frac{t}{R}\lambda \right)} \sum_{k=0}^{\infty} \frac{1}{k!} \sum_{\kappa} \int_{\mathbf{A} > \mathbf{0}} |\mathbf{A}|^{a-(m+1)/2} \\ &\quad \times \text{etr} \left(-R\mathbf{A} \left(1 - \frac{t}{R}\lambda \right) \lambda^{-1} \right) \\ &\quad \times \text{tr} \left(R\mathbf{A} \left(1 - \frac{t}{R}\lambda \right) \lambda^{-1} \right) C_{\kappa} (R\mathbf{A}\lambda^{-1} (\mathbf{I} - \lambda\Sigma^{-1})) (d\mathbf{A}). \quad (2.3) \end{aligned}$$

This integral needs to be studied in detail. Let us first start with the integral [Caro-Lopera et al. (2009, Theorem 1)]

$$\begin{aligned} &\int_{\mathbf{X} > \mathbf{0}} h(\text{tr } \mathbf{XZ}) |\mathbf{X}|^{a-(m+1)/2} C_{\kappa}(\mathbf{XY})(d\mathbf{X}) \\ &= \frac{\Gamma_m(a)}{\Gamma(ma+k)} |\mathbf{Z}|^{-a}(a)_{\kappa} C_{\kappa}(\mathbf{YZ}^{-1}) \int_0^{\infty} h(w) w^{ma+k-1} dw, \quad (2.4) \end{aligned}$$

where the univariate integral is finite for the particular generator function $h(\cdot)$, $\mathbf{Z}$ is a complex symmetric $m \times m$ matrix with $\Re(\mathbf{Z}) > \mathbf{0}$, $\mathbf{Y}$ is a symmetric $m \times m$ matrix, and κ is an ordered partition of k , as mentioned earlier. We now generalize a result of Khatri (1966), which was established by expanding a Laplace transform; here, it is deduced simply by taking $h(y) = y^j e^{-y}$ in (2.4).

LEMMA 2.1.

$$\begin{aligned} &\int_{\mathbf{X} > \mathbf{0}} \text{etr}(-\mathbf{XZ})(\text{tr}(\mathbf{XZ}))^j |\mathbf{X}|^{a-(m+1)/2} C_{\kappa}(\mathbf{XY})(d\mathbf{X}) \\ &= \frac{\Gamma_m(a)\Gamma(ma+j+k)}{\Gamma(ma+k)} |\mathbf{Z}|^{-a}(a)_{\kappa} C_{\kappa}(\mathbf{YZ}^{-1}). \end{aligned}$$

Applying Lemma 2.1 to (2.3), we obtain, after some algebraic simplification, that

$$S = |\lambda^{-1}\Sigma|^{-a} \sum_{k=0}^{\infty} \frac{(ma+k)t\lambda}{k!R \left(1 - \frac{t}{R}\lambda\right)^{k+ma+1}} \sum_{\kappa} (a)_{\kappa} C_{\kappa}(\mathbf{I} - \lambda\Sigma^{-1}),$$

which can be expressed as

$$S = |\lambda^{-1}\Sigma|^{-a} \left[\sum_{k=0}^{\infty} \frac{(ma+k)}{k! \left(1 - \frac{t}{R}\lambda\right)^{k+ma+1}} \sum_{\kappa} (a)_{\kappa} C_{\kappa}(\mathbf{I} - \lambda\Sigma^{-1}) - \sum_{k=0}^{\infty} \frac{(ma+k)}{k! \left(1 - \frac{t}{R}\lambda\right)^{k+ma}} \sum_{\kappa} (a)_{\kappa} C_{\kappa}(\mathbf{I} - \lambda\Sigma^{-1}) \right].$$

The moment generating function in (2.2) is a sum indexed by the moment generating functions of chi-squared distributions. Explicitly, we have

$$\begin{aligned} \phi(t) = |\lambda^{-1}\Sigma|^{-\frac{n}{2}} & \left[\sum_{k=0}^{\infty} \frac{1}{k! \left(1 - \frac{t}{R}\lambda\right)^{k+\frac{mn}{2}}} \sum_{\kappa} \left(\frac{n}{2}\right)_{\kappa} C_{\kappa}(\mathbf{I} - \lambda\Sigma^{-1}) \right. \\ & + \sum_{k=0}^{\infty} \frac{\left(\frac{mn}{2} + k\right)}{k! \frac{mn}{2} \left(1 - \frac{t}{R}\lambda\right)^{k+\frac{mn}{2}+1}} \sum_{\kappa} \left(\frac{n}{2}\right)_{\kappa} C_{\kappa}(\mathbf{I} - \lambda\Sigma^{-1}) \\ & \left. - \sum_{k=0}^{\infty} \frac{\left(\frac{mn}{2} + k\right)}{k! \frac{mn}{2} \left(1 - \frac{t}{R}\lambda\right)^{k+\frac{mn}{2}}} \sum_{\kappa} \left(\frac{n}{2}\right)_{\kappa} C_{\kappa}(\mathbf{I} - \lambda\Sigma^{-1}) \right]. \end{aligned}$$

Upon inverting term by term, we obtain the density function of $\text{tr } \mathbf{A}$ for the Kotz model with generator function $h(y) = 2(mn)^{-1}\pi^{-mn/2}R^{1+mn/2}y e^{-Ry}$, ($R, y > 0$), as

$$\begin{aligned} f_{\text{tr } \mathbf{A}}(u) = |\lambda^{-1}\Sigma|^{-\frac{n}{2}} & \left[\sum_{k=0}^{\infty} \frac{e^{-uR/\lambda} u^{k+\frac{mn}{2}-1}}{k! \left(\frac{\lambda}{R}\right)^{k+\frac{mn}{2}} \Gamma\left(k + \frac{mn}{2}\right)} \sum_{\kappa} \left(\frac{n}{2}\right)_{\kappa} C_{\kappa}(\mathbf{I} - \lambda\Sigma^{-1}) \right. \\ & + \sum_{k=0}^{\infty} \frac{\left(\frac{mn}{2} + k\right) e^{-uR/\lambda} u^{k+\frac{mn}{2}}}{k! \frac{mn}{2} \left(\frac{\lambda}{R}\right)^{k+\frac{mn}{2}+1} \Gamma\left(k + \frac{mn}{2} + 1\right)} \\ & \left. \times \sum_{\kappa} \left(\frac{n}{2}\right)_{\kappa} C_{\kappa}(\mathbf{I} - \lambda\Sigma^{-1}) \right] \end{aligned}$$

$$\begin{aligned}
 & - \sum_{k=0}^{\infty} \frac{\left(\frac{mn}{2} + k\right) e^{-uR/\lambda} u^{k + \frac{mn}{2} - 1}}{k! \frac{mn}{2} \left(\frac{\lambda}{R}\right)^{k + \frac{mn}{2}} \Gamma\left(k + \frac{mn}{2}\right)} \\
 & \times \sum_{\kappa} \left(\frac{n}{2}\right)_{\kappa} C_{\kappa}(\mathbf{I} - \lambda \Sigma^{-1}) \Bigg], u > 0.
 \end{aligned}$$

We have thus established the following result.

COROLLARY 2.2. *If $\mathbf{A}$ is $EW_m(n, \Sigma, h)$ with $n \geq m$ and*

$$h(y) = 2(mn)^{-1} \pi^{-\frac{mn}{2}} R^{1 + \frac{mn}{2}} y e^{-Ry}, y > 0,$$

then the distribution function of $\text{tr } \mathbf{A}$ can be expressed as

$$\begin{aligned}
 P(\text{tr } \mathbf{A} \leq x) &= |\lambda^{-1} \Sigma|^{-\frac{n}{2}} \left[\sum_{k=0}^{\infty} \frac{1}{k!} P\left(\chi_{mn+2k}^2 \leq \frac{x}{\frac{\lambda}{2R}}\right) \sum_{\kappa} \left(\frac{n}{2}\right)_{\kappa} C_{\kappa}(\mathbf{I} - \lambda \Sigma^{-1}) \right. \\
 &+ \sum_{k=0}^{\infty} \frac{\left(\frac{mn}{2} + k\right)}{k! \frac{mn}{2}} P\left(\chi_{mn+2k+2}^2 \leq \frac{x}{\frac{\lambda}{2R}}\right) \\
 &\times \sum_{\kappa} \left(\frac{n}{2}\right)_{\kappa} C_{\kappa}(\mathbf{I} - \lambda \Sigma^{-1}) \\
 &- \sum_{k=0}^{\infty} \frac{\left(\frac{mn}{2} + k\right)}{k! \frac{mn}{2}} P\left(\chi_{mn+2k}^2 \leq \frac{x}{\frac{\lambda}{2R}}\right) \\
 &\left. \times \sum_{\kappa} \left(\frac{n}{2}\right)_{\kappa} C_{\kappa}(\mathbf{I} - \lambda \Sigma^{-1}) \right].
 \end{aligned}$$

As in the Gaussian case, the parameter λ is chosen so that the series converges rapidly. Khatri and Srivastava (1971) suggested the choice of $\lambda = 2\lambda_1\lambda_2/\lambda_1 + \lambda_2$, where λ_1 and λ_2 are the largest and smallest latent roots of Σ , respectively.

It should be mentioned here that other elliptical models, such as Pearson type, Bessel and Jensen-logistic, can be studied in an analogous manner by using the developed approach.

3 The Sphericity Test

We can obtain the s -th moment of the sphericity test statistic V for a general elliptical model with generator function $h(\cdot)$, as presented in the following theorem.

THEOREM 3.1. *If*

$$V = \frac{|\mathbf{A}|}{\left[\frac{1}{m} \operatorname{tr} \mathbf{A}\right]^m},$$

where $\mathbf{A}$ is $EW_m(n, \Sigma, h)$ with $n \geq m$, then

$$E(V^s) = \frac{m^{ms} \Gamma_m\left(\frac{n}{2} + s\right)}{\pi^{ms} \Gamma_m\left(\frac{n}{2}\right)} |\Sigma|^s E((\operatorname{tr} \mathbf{A})^{-ms});$$

here, $E((\operatorname{tr} \mathbf{A})^{-ms})$ is the $(-ms)$ th moment of $\operatorname{tr} \mathbf{A}$ computed from the density

$$f_{\operatorname{tr} \mathbf{A}}(u) = \pi^{m(n/2+s)} \sum_{k=0}^{\infty} \frac{1}{k!} r(u, k) du,$$

where the density function $r(u, k)$ is obtained by inverting the moment generating function

$$M_u(t, k, h) = \frac{\int_0^{\infty} h(w) w^{\frac{mn}{2} + ms + k - 1} dw}{\Gamma\left(\frac{mn}{2} + ms + k\right)} \sum_{\kappa} \left(\frac{n}{2} + s\right)_{\kappa} C_{\kappa}(t\Sigma).$$

PROOF. We have

$$\begin{aligned} E(V^s) &= \frac{m^{ms} \pi^{mn/2}}{\Gamma_m\left(\frac{n}{2}\right) |\Sigma|^{n/2}} \int_{\mathbf{A} > \mathbf{0}} h(\operatorname{tr} \Sigma^{-1} \mathbf{A}) |\mathbf{A}|^{(n+2s-m-1)/2} (\operatorname{tr} \mathbf{A})^{-ms} (d\mathbf{A}) \\ &= \frac{m^{ms} \Gamma_m\left(\frac{n}{2} + s\right)}{\pi^{ms} \Gamma_m\left(\frac{n}{2}\right)} |\Sigma|^s E((\operatorname{tr} \mathbf{A})^{-ms}), \end{aligned} \quad (3.1)$$

where in obtaining the last equation we have used the fact that

$$E((\operatorname{tr} \mathbf{A})^{-ms}) = \frac{\pi^{m(n/2+s)} \int_{\mathbf{A} > \mathbf{0}} h(\operatorname{tr} \Sigma^{-1} \mathbf{A}) |\mathbf{A}|^{(n+2s-m-1)/2} (\operatorname{tr} \mathbf{A})^{-ms} (d\mathbf{A})}{\Gamma_m\left(\frac{n}{2} + s\right) |\Sigma|^{\frac{n}{2} + s}}$$

when $\mathbf{A}$ is $EW_m(n + 2s, \Sigma, h)$.

Thus, in this case, we have

$$f_{\operatorname{tr} \mathbf{A}}(u) = \pi^{m(n/2+s)} \sum_{k=0}^{\infty} \frac{1}{k!} r(u, k) du.$$

Here again, the density function $r(u, k)$ is obtained by inverting the moment generating function

$$M_u(t, k, h) = \frac{\int_0^{\infty} h(w) w^{\frac{mn}{2} + ms + k - 1} dw}{\Gamma\left(\frac{mn}{2} + ms + k\right)} \sum_{\kappa} \left(\frac{n}{2} + s\right)_{\kappa} C_{\kappa}(t\Sigma).$$

Finally, $E((\text{tr } \mathbf{A})^{-ms})$ is obtained by computing the $(-ms)$ th moment from $r(u, k)$.

It is clear that for a specific elliptical model, the preceding expression can be derived explicitly. Moreover, the expressions require only the evaluation of a simple univariate integral involving the generator function, instead of the computation of the usual multivariate integrals. For example, we derive the expression for the Gaussian case [studied earlier by Khatri and Srivastava (1971) and Muirhead (1982)] and the non-normal model with generator function $h(y) = e^{-Ry} y$, $R > 0$.

COROLLARY 3.1. *If*

$$V = \frac{|\mathbf{A}|}{\left[\frac{1}{m} \text{tr } \mathbf{A}\right]^m},$$

where $\mathbf{A}$ is $EW_m(n, \Sigma, h)$ with $n \geq m$, then

$$\begin{aligned} E(V^s) &= \frac{m^{ms} \Gamma(mn/2) \Gamma_m\left(\frac{n}{2} + s\right)}{\Gamma(mn/2 + ms) \Gamma_m\left(\frac{n}{2}\right)} |\lambda \Sigma^{-1}|^{n/2} \\ &\quad \times \sum_{k=0}^{\infty} \frac{(mn/2)_k}{k! (mn/2 + ms)_k} \times \sum_{\kappa} (n/2 + s)_{\kappa} C_{\kappa}(\mathbf{I} - \lambda \Sigma^{-1}), \end{aligned}$$

with $0 < \lambda < \infty$ being arbitrary.

PROOF. Starting from Corollary 2.1, we have

$$\begin{aligned} f_{\text{tr } \mathbf{A}}(u) &= \pi^{m(n/2+s)} \sum_{k=0}^{\infty} \frac{1}{k!} r(u, k) du \\ &= |\lambda^{-1} \Sigma|^{-n/2-s} \sum_{k=0}^{\infty} \frac{e^{-u/2\lambda} u^{mn/2+ms+k}}{k! (2\lambda)^{mn/2+ms+k} \Gamma(mn/2 + ms + k)} \\ &\quad \times \sum_{\kappa} (n/2 + s)_{\kappa} C_{\kappa}(\mathbf{I} - \lambda \Sigma^{-1}), \end{aligned}$$

from which we obtain

$$\begin{aligned} E((\text{tr } \mathbf{A})^{-ms}) &= |\lambda^{-1} \Sigma|^{-n/2-s} (2\lambda)^{-ms} \sum_{k=0}^{\infty} \frac{\Gamma(mn/2 + k)}{k! \Gamma(mn/2 + ms + k)} \\ &\quad \times \sum_{\kappa} (n/2 + s)_{\kappa} C_{\kappa}(\mathbf{I} - \lambda \Sigma^{-1}). \end{aligned}$$

The required result then follows readily by (3.1).

The second example considers the non-Gaussian model in Corollary 2.2 for which the moments are obtained in the following way.

COROLLARY 3.2. *If $V = |\mathbf{A}|/[1/m \operatorname{tr} \mathbf{A}]^m$, $\mathbf{A}$ is $EW_m(n, \mathbf{\Sigma}, h)$ with $n \geq m$, and $h(y) = 2(mn)^{-1} \pi^{-mn/2} R^{1+mn/2} y e^{-Ry}$, $R > 0$, then*

$$\begin{aligned} E[V^s] &= \frac{m^{ms} \Gamma_m(\frac{1}{2}n + s)}{\frac{mn}{2} \Gamma_m(\frac{1}{2}n)} |\lambda \mathbf{\Sigma}^{-1}|^{\frac{n}{2}} \sum_{k=0}^{\infty} \sum_{\kappa} \left(\frac{n}{2} + s\right)_{\kappa} C_{\kappa}(\mathbf{I} - \lambda \mathbf{\Sigma}^{-1}) \\ &\quad \times \left[\frac{\Gamma(k + \frac{mn}{2} + 1) (\frac{mn}{2} + ms)}{k! \Gamma(k + \frac{mn}{2} + ms + 1)} + \frac{\Gamma(k + \frac{mn}{2} + 2)}{k! \Gamma(k + \frac{mn}{2} + ms + 2)} \right. \\ &\quad \left. - \frac{\Gamma(k + \frac{mn}{2} + 1)}{k! \Gamma(k + \frac{mn}{2} + ms + 1)} \right], \end{aligned}$$

where $0 < \lambda < \infty$ is arbitrary.

PROOF. First, note that if $\mathbf{A}$ is $EW_m(n, \mathbf{\Sigma}, h)$, $n \geq m$, and $h(y) = 2(mn)^{-1} \pi^{-mn/2} R^{1+mn/2} y e^{-Ry}$, then

$$E[V^s] = \frac{(R^{-1}m)^{ms} \Gamma_m(\frac{1}{2}n + s) (1 + \frac{2s}{n})}{\Gamma_m(\frac{1}{2}n) |\mathbf{\Sigma}|^{-s}} E[(\operatorname{tr} \mathbf{A})^{-ms}],$$

where $\mathbf{A}$ is $EW_m(r, \mathbf{\Sigma}, h)$ with $r \geq m$, $h(y) = 2(mr)^{-1} \pi^{-mr/2} R^{1+mr/2} y e^{-Ry}$, and $r = n + 2s$; see (3.1). In this case, the density function of $\operatorname{tr} \mathbf{A}$ is given by Corollary 2.2, with $n = n + 2s$, as

$$\begin{aligned} f_{\operatorname{tr} \mathbf{A}}(u) &= |\lambda^{-1} \mathbf{\Sigma}|^{-\frac{n}{2}-s} \\ &\quad \times \left[\sum_{k=0}^{\infty} \frac{e^{-uR/\lambda} u^{k + \frac{mn}{2} + ms - 1}}{k! \left(\frac{\lambda}{R}\right)^{k + ms + \frac{mn}{2}} \Gamma(k + ms + \frac{mn}{2})} \right. \\ &\quad \times \sum_{\kappa} \left(\frac{n}{2} + s\right)_{\kappa} C_{\kappa}(\mathbf{I} - \lambda \mathbf{\Sigma}^{-1}) \\ &\quad + \sum_{k=0}^{\infty} \frac{(\frac{mn}{2} + ms + k) e^{-uR/\lambda} u^{k + ms + \frac{mn}{2}}}{k! (\frac{mn}{2} + ms) \left(\frac{\lambda}{R}\right)^{k + \frac{mn}{2} + ms + 1} \Gamma(k + \frac{mn}{2} + ms + 1)} \\ &\quad \left. \times \sum_{\kappa} \left(\frac{n}{2} + s\right)_{\kappa} C_{\kappa}(\mathbf{I} - \lambda \mathbf{\Sigma}^{-1}) \right] \end{aligned}$$

$$\begin{aligned}
 & - \sum_{k=0}^{\infty} \frac{\left(\frac{mn}{2} + ms + k\right) e^{-uR/\lambda} u^{k + \frac{mn}{2} + ms - 1}}{k! \left(\frac{mn}{2} + ms\right) \left(\frac{\lambda}{R}\right)^{k + \frac{mn}{2} + ms} \Gamma\left(k + \frac{mn}{2} + ms\right)} \\
 & \times \sum_{\kappa} \left(\frac{n}{2} + s\right)_{\kappa} C_{\kappa}(\mathbf{I} - \lambda \mathbf{\Sigma}^{-1}) \Bigg].
 \end{aligned}$$

Thus, by using the fact that the $(-ms)th$ moment of the gamma distribution with parameters a and b is $b^{-ms} \Gamma(a - ms) / \Gamma(a)$, we obtain from the above density function that

$$\begin{aligned}
 E((\text{tr } \mathbf{A})^{-ms}) &= |\lambda^{-1} \mathbf{\Sigma}|^{-\frac{n}{2} - s} \left(\frac{\lambda}{R}\right)^{-ms} \\
 & \times \left[\sum_{k=0}^{\infty} \frac{\Gamma\left(k + \frac{mn}{2}\right)}{k! \Gamma\left(k + ms + \frac{mn}{2}\right)} \sum_{\kappa} \left(\frac{n}{2} + s\right)_{\kappa} C_{\kappa}(\mathbf{I} - \lambda \mathbf{\Sigma}^{-1}) \right. \\
 & + \sum_{k=0}^{\infty} \frac{\left(\frac{mn}{2} + ms + k\right) \Gamma\left(k + \frac{mn}{2} + 1\right)}{k! \left(\frac{mn}{2} + ms\right) \Gamma\left(k + \frac{mn}{2} + ms + 1\right)} \\
 & \times \sum_{\kappa} \left(\frac{n}{2} + s\right)_{\kappa} C_{\kappa}(\mathbf{I} - \lambda \mathbf{\Sigma}^{-1}) \\
 & - \sum_{k=0}^{\infty} \frac{\left(\frac{mn}{2} + ms + k\right) \Gamma\left(k + \frac{mn}{2}\right)}{k! \left(\frac{mn}{2} + ms\right) \Gamma\left(k + ms + \frac{mn}{2}\right)} \\
 & \left. \times \sum_{\kappa} \left(\frac{n}{2} + s\right)_{\kappa} C_{\kappa}(\mathbf{I} - \lambda \mathbf{\Sigma}^{-1}) \right],
 \end{aligned}$$

which is the required result.

Comparing the above result for $E[V^s]$ with the corresponding expression for the Gaussian case in Corollary 3.1, we see that the moments of the two models are quite different.

Now, we can state the general result corresponding to the Kotz model.

COROLLARY 3.3. *If $\mathbf{A}$ is $EW_m(n, \mathbf{\Sigma}, h)$ with $n \geq m$, and*

$$h(y) = \frac{R^{\frac{2T+mn-2}{2}} \Gamma\left(\frac{mn}{2}\right)}{\pi^{\frac{mn}{2}} \Gamma\left(\frac{2T+mn-2}{2}\right)} y^{T-1} e^{-Ry}, \quad R > 0, \quad T \in N,$$

then, for suitable coefficients $\beta_{j,i}^{i,j}$, α_i and $\delta_{r,j}$, we have

$$P(\text{tr } \mathbf{A} \leq x) = |\lambda^{-1} \mathbf{\Sigma}|^{-a} \sum_{j=0}^{T-1} \left(\sum_{i=j}^{T-1} \beta_{j,i}^{i,j} \alpha_i \right) \sum_{k=0} \frac{(ma+k)_j}{k!} \\ \times \sum_{r=0}^j \delta_{r,j} P \left(\chi_{mn+2k+2r}^2 \leq \frac{x}{\frac{\lambda}{2R}} \right) \sum_{\kappa} (a)_{\kappa} C_{\kappa}(\mathbf{I} - \lambda \mathbf{\Sigma}^{-1}).$$

PROOF. Consider the Kotz type model with generator function

$$h(y) = \frac{rR^{\frac{T+ma-1}{r}} \Gamma(ma)}{\pi^{ma} \Gamma\left(\frac{T+ma-1}{r}\right)} y^{T-1} e^{-Ry^r}, \quad y > 0,$$

where $a = n/2$. After some simplification, we have

$$\phi(t) = \pi^{mn/2} \sum_{k=0}^{\infty} \frac{\int_0^{\infty} h(w) w^{\frac{mn}{2}+k-1} dw}{k! \Gamma\left(\frac{mn}{2} + k\right)} \sum_{\kappa} \left(\frac{n}{2}\right)_{\kappa} C_{\kappa}(t \mathbf{\Sigma}) \\ = \Gamma(ma) \sum_{k=0}^{\infty} \frac{\left(\frac{T+ma-1}{r}\right)_{\frac{k}{r}}}{k! \Gamma(ma+k)} \sum_{\kappa} \left(\frac{n}{2}\right)_{\kappa} C_{\kappa}(tR^{-1/r} \mathbf{\Sigma}).$$

An important simplification is obtained for the case $r = 1$ and $T \in N$, in terms of the Pochhammer symbol (a polynomial in k of degree $T - 1$), as follows:

$$\phi(t) = \sum_{k=0}^{\infty} \frac{(T+ma-1)_k}{k! (ma)_k} \sum_{\kappa} \left(\frac{n}{2}\right)_{\kappa} C_{\kappa}(tR^{-1} \mathbf{\Sigma}) \\ = \sum_{k=0}^{\infty} \frac{(k+ma)_{T-1}}{k! (ma)_{T-1}} \sum_{\kappa} \left(\frac{n}{2}\right)_{\kappa} C_{\kappa}(tR^{-1} \mathbf{\Sigma}). \quad (3.2)$$

Now, it is clear that there exist positive integers $\beta_{j,i}^{i,j}$'s such that the following zonal polynomial expansion holds:

$$\left[\alpha_0 + \left(\sum_{i=1}^{T-1} \beta_{1,i}^{i,1} \alpha_i \right) \text{tr}(\mathbf{XY}) + \left(\sum_{i=2}^{T-1} \beta_{2,i}^{i,2} \alpha_i \right) \text{tr}^2(\mathbf{XY}) + \cdots + \right. \\ \left. + \left(\sum_{i=T-1-1}^{T-1} \beta_{T-1-1,i}^{i,T-1-1} \alpha_i \right) \text{tr}^{T-1-1}(\mathbf{XY}) + \alpha_{T-1} \text{tr}^{T-1}(\mathbf{XY}) \right] \text{etr}(\mathbf{XY})$$

$$\begin{aligned}
 &= \text{etr}(\mathbf{XY}) \sum_{j=0}^{T-1} \left(\sum_{i=j}^{T-1} \beta_{j,i}^{i,j} \alpha_i \right) \text{tr}^j(\mathbf{XY}) \\
 &= \sum_{k=0}^{\infty} \frac{\alpha_0 + \alpha_1 k + \cdots + \alpha_{T-1} k^{T-1}}{k!} [\text{tr}(\mathbf{XY})]^k \\
 &= \sum_{k=0}^{\infty} \frac{\alpha_0 + \alpha_1 k + \cdots + \alpha_{T-1} k^{T-1}}{k!} \sum_{\tau} C_{\tau}(\mathbf{XY}). \tag{3.3}
 \end{aligned}$$

The construction of the coefficients α 's and β 's is out of the scope of this work, and is planned to be taken up in an independent study.

An integral representation of (3.2) follows from (3.3), as

$$\begin{aligned}
 S &:= \sum_{k=0}^{\infty} \frac{(k+ma)_{T-1}}{k!(ma)_{T-1}} \sum_{\kappa} \left(\frac{n}{2} \right)_{\kappa} C_{\kappa}(tR^{-1}\Sigma) \\
 &= \frac{|R\Sigma^{-1}|^a}{\Gamma_m(a)} \sum_{j=0}^{T-1} \left(\sum_{i=j}^{T-1} \beta_{j,i}^{i,j} \alpha_i \right) \\
 &\quad \times \int_{\mathbf{A} > \mathbf{0}} \text{etr}(-\mathbf{A}R\Sigma^{-1}) |\mathbf{A}|^{a-(m+1)/2} \text{etr}(t\mathbf{A}) \text{tr}^j(t\mathbf{A}) (d\mathbf{A}).
 \end{aligned}$$

For an arbitrary $0 < \lambda < \infty$, we have

$$\begin{aligned}
 S &= \frac{|R\Sigma^{-1}|^a}{\Gamma_m(a)} \sum_{j=0}^{T-1} \left(\sum_{i=j}^{T-1} \beta_{j,i}^{i,j} \alpha_i \right) t^j \\
 &\quad \times \int_{\mathbf{A} > \mathbf{0}} |\mathbf{A}|^{a-(m+1)/2} \text{etr} \left(-R\mathbf{A} \left(1 - \frac{t}{R} \lambda \right) \lambda^{-1} \right) \text{tr}^j(\mathbf{A}) \\
 &\quad \times \text{etr} \left(R\mathbf{A} \lambda^{-1} (\mathbf{I} - \lambda \Sigma^{-1}) \right) (d\mathbf{A}).
 \end{aligned}$$

Upon expanding in terms of zonal polynomials, we obtain

$$\begin{aligned}
 S &= \frac{|R\Sigma^{-1}|^a}{\Gamma_m(a)} \sum_{j=0}^{T-1} \left(\sum_{i=j}^{T-1} \beta_{j,i}^{i,j} \alpha_i \right) \frac{t^j \lambda^j}{R^j (1 - \frac{t}{R} \lambda)^j} \\
 &\quad \times \sum_{k=0}^{\infty} \frac{1}{k!} \sum_{\kappa} \int_{\mathbf{A} > \mathbf{0}} |\mathbf{A}|^{a-(m+1)/2} \text{etr} \left(-R\mathbf{A} \left(1 - \frac{t}{R} \lambda \right) \lambda^{-1} \right) \\
 &\quad \times \text{tr}^j \left(R\mathbf{A} \left(1 - \frac{t}{R} \lambda \right) \lambda^{-1} \right) C_{\kappa} (R\mathbf{A} \lambda^{-1} (\mathbf{I} - \lambda \Sigma^{-1})) (d\mathbf{A}).
 \end{aligned}$$

Now, we can use Lemma 2.1 to obtain

$$S = |\lambda^{-1}\Sigma|^{-a} \sum_{j=0}^{T-1} \left(\sum_{i=j}^{T-1} \beta_{j,i}^{i,j} \alpha_i \right) \sum_{k=0} \frac{R^{-j}(ma+k)_j t^j \lambda^j}{k! \left(1 - \frac{t}{R} \lambda\right)^{ma+k+j}} \\ \times \sum_{\kappa} (a)_{\kappa} C_{\kappa}(\mathbf{I} - \lambda \Sigma^{-1}).$$

For any $j = 0, 1, 2, \dots$, there exist coefficients $\delta_{r,j}$ such that

$$S = |\lambda^{-1}\Sigma|^{-a} \sum_{j=0}^{T-1} \left(\sum_{i=j}^{T-1} \beta_{j,i}^{i,j} \alpha_i \right) \sum_{k=0} \frac{(ma+k)_j}{k!} \\ \times \sum_{r=0}^j \frac{\delta_{r,j}}{\left(1 - \frac{t}{R} \lambda\right)^{ma+k+r}} \sum_{\kappa} (a)_{\kappa} C_{\kappa}(\mathbf{I} - \lambda \Sigma^{-1}).$$

Using the moments of a univariate gamma distribution, we then obtain the density function of $\text{tr } \mathbf{A}$ for the non-Gaussian Kotz model with generator function $h(y) = R^{T+mn/2-1} \Gamma(mn/2) / \pi^{mn/2} \Gamma(T+mn/2-1) y^{T-1} e^{-Ry}$ with $y > 0$, $T \in \mathbb{Z}^+$, $R > 0$, as

$$f_{\text{tr } \mathbf{A}}(u) = |\lambda^{-1}\Sigma|^{-a} \sum_{j=0}^{T-1} \left(\sum_{i=j}^{T-1} \beta_{j,i}^{i,j} \alpha_i \right) \sum_{k=0} \frac{(ma+k)_j}{k!} \\ \times \sum_{r=0}^j \frac{\delta_{r,j}}{\left(\frac{\lambda}{R}\right)^{ma+k+r}} \frac{e^{-uR/\lambda} u^{ma+k+r-1}}{\Gamma(ma+k+r)} \sum_{\kappa} (a)_{\kappa} C_{\kappa}(\mathbf{I} - \lambda \Sigma^{-1}), u > 0.$$

Integrating the above expression from 0 to x , we obtain the cumulative distribution function of $\text{tr } \mathbf{A}$ as

$$P(\text{tr } \mathbf{A} \leq x) = |\lambda^{-1}\Sigma|^{-a} \sum_{j=0}^{T-1} \left(\sum_{i=j}^{T-1} \beta_{j,i}^{i,j} \alpha_i \right) \sum_{k=0} \frac{(ma+k)_j}{k!} \\ \times \sum_{r=0}^j \delta_{r,j} P\left(\chi_{mn+2k+2r}^2 \leq \frac{x}{\frac{\lambda}{2R}}\right) \sum_{\kappa} (a)_{\kappa} C_{\kappa}(\mathbf{I} - \lambda \Sigma^{-1}),$$

which is the required result.

The results of Khatri and Srivastava (1971) and Muirhead (1982) for the Gaussian case follow immediately by taking $T = 1$ and $j = 0$ in the above result.

We now present expressions for the moments of a generalized modified likelihood ratio statistic under the Kotz model.

COROLLARY 3.4. *If $V = |\mathbf{A}|/[1/m \operatorname{tr} \mathbf{A}]^m$, $\mathbf{A}$ is $EW_m(n, \Sigma, h)$ with $n \geq m$, and*

$$h(y) = \frac{R^{T+\frac{mn}{2}-1} \Gamma\left(\frac{mn}{2}\right)}{\pi^{\frac{mn}{2}} \Gamma\left(T + \frac{mn}{2} - 1\right)} y^{T-1} e^{-Ry}, \quad y > 0, R > 0, T \in \mathbb{N},$$

then

$$\begin{aligned} E[V^s] &= \frac{m^{ms} \Gamma\left(\frac{1}{2}n + s\right) |\lambda \Sigma^{-1}|^{n/2} \left(\frac{mn}{2} + ms\right)_{T-1}}{\Gamma_m\left(\frac{1}{2}n\right) \left(\frac{mn}{2}\right)_{T-1}} \\ &\quad \times \sum_{j=0}^{T-1} \left(\sum_{i=j}^{T-1} \beta_{j,i}^{i,j} \alpha_i \right) \sum_{k=0} \frac{\left(\frac{mn}{2} + ms + k\right)_j}{k!} \\ &\quad \times \sum_{r=0}^j \frac{\delta_{r,j} \Gamma(mn/2 + k + r)}{\Gamma(mn/2 + k + r + ms)} \sum_{\kappa} (n/2 + s)_{\kappa} C_{\kappa}(\mathbf{I} - \lambda \Sigma^{-1}), \end{aligned}$$

where $0 < \lambda < \infty$ is arbitrary.

PROOF. Let $\mathbf{A}$ be $EW_m(n, \Sigma, h)$ with $n \geq m$, and the generator function h be

$$h(y) = \frac{R^{T-1+\frac{mn}{2}} \Gamma\left(\frac{mn}{2}\right)}{\pi^{\frac{mn}{2}} \Gamma\left(T - 1 + \frac{mn}{2}\right)} y^{T-1} e^{-Ry}, \quad y > 0, R > 0, T \in \mathbb{N}.$$

Then,

$$E[V^s] = \frac{m^{ms} \Gamma\left(\frac{1}{2}n + s\right) |\Sigma|^s \left(\frac{mn}{2} + ms\right)_{T-1}}{R^{ms} \Gamma_m\left(\frac{1}{2}n\right) \left(\frac{mn}{2}\right)_{T-1}} E[(\operatorname{tr} \mathbf{A})^{-ms}].$$

Here, $\mathbf{A}$ is $EW_m(n, \Sigma, h)$ with $n \geq m$ and $n = n + 2s$; see (3.1).

So, the density function of $\operatorname{tr} \mathbf{A}$ is given by Corollary 2.2 with $n = n + 2s$ as

$$\begin{aligned} f_{\operatorname{tr} \mathbf{A}}(u) &= |\lambda^{-1} \Sigma|^{-n/2-s} \sum_{j=0}^{T-1} \left(\sum_{i=j}^{T-1} \beta_{j,i}^{i,j} \alpha_i \right) \sum_{k=0} \frac{\left(\frac{mn}{2} + ms + k\right)_j}{k!} \\ &\quad \times \sum_{r=0}^j \frac{\delta_{r,j} e^{-u/(\lambda/R)} u^{ma+k+r+ms-1}}{(\lambda/R)^{ma+k+r+ms} \Gamma(ma + k + r + ms)} \\ &\quad \times \sum_{\kappa} (a + s)_{\kappa} C_{\kappa}(\mathbf{I} - \lambda \Sigma^{-1}). \end{aligned}$$

Thus, we have

$$\begin{aligned}
 E((\text{tr } \mathbf{A})^{-ms}) &= \frac{|\lambda^{-1} \mathbf{\Sigma}|^{-n/2-s}}{(\lambda/R)^{ms}} \sum_{j=0}^{T-1} \left(\sum_{i=j}^{T-1} \beta_{j,i}^{i,j} \alpha_i \right) \sum_{k=0}^{\left(\frac{mn}{2} + ms + k\right)_j} \frac{1}{k!} \\
 &\times \sum_{r=0}^j \frac{\delta_{r,j} \Gamma(mn/2 + k + r)}{\Gamma(mn/2 + k + r + ms)} \sum_{\kappa} (n/2 + s)_{\kappa} C_{\kappa}(\mathbf{I} - \lambda \mathbf{\Sigma}^{-1}),
 \end{aligned}$$

which is the required result.

It should be mentioned that the moments derived here can be quickly and efficiently computed by using the algorithm of Koev and Edelman (2006). For the Kotz model, we have the same structure as in the normal case, and they preserve the conditions on λ so as to obtain a fast convergence of the involved series. Moreover, the asymptotic non-null and null distributions of the likelihood ratio statistic can be obtained by a suitable change in the procedure for the Gaussian case presented by Muirhead (1982). Finally, the moments of the test statistics V may be used to obtain an exact expression for the density function by employing the Mellin transform approach.

4 Some Exact Non-Null Distributions of the V -Statistic

Now for illustrating the developed technique, we derive the exact distribution of the V -statistic for the case of the Kotz family, by using the general moments derived in Corollary 3.4. First, we start with the multivariate Gaussian case.

As usual, we can write the moments $E(V^s)$ in terms of the Mellin transform $M(s+1)$, and then invert it to obtain the density function $f(W)$; that is,

$$E(W^s) = M(s+1) = \int_0^{\infty} W^s f(W) dW.$$

In the multivariate Gaussian case, we have

$$\begin{aligned}
 E(V^s) &= \frac{m^{ms} \Gamma(mn/2) \Gamma_m\left(\frac{n}{2} + s\right)}{\Gamma(mn/2 + ms) \Gamma_m\left(\frac{n}{2}\right)} |\lambda \mathbf{\Sigma}^{-1}|^{n/2} \\
 &\times \sum_{k=0}^{\infty} \frac{(mn/2)_k}{k! (mn/2 + ms)_k} \times \sum_{\kappa} (n/2 + s)_{\kappa} C_{\kappa}(\mathbf{I} - \lambda \mathbf{\Sigma}^{-1}),
 \end{aligned}$$

where $0 < \lambda < \infty$ is arbitrary. Therefore, the required density in this case is given by

$$\begin{aligned}
 f(v) &= (2\pi i)^{-1} \int_{c-i\infty}^{c+i\infty} v^{-s-1} E(v^s) ds \\
 &= \frac{\pi^{\frac{1}{4}m(m-1)} |\lambda \Sigma^{-1}|^{\frac{1}{2}n} (2\pi)^{\frac{1}{2}(m-1)}}{\Gamma_m\left(\frac{1}{2}n\right) v} \\
 &\quad \times \sum_{k=0}^{\infty} \sum_{\kappa} \frac{1}{k!} m^{\frac{1}{2}-\frac{1}{2}mn-k} \Gamma\left(\frac{1}{2}mn+k\right) C_{\kappa}(\mathbf{I} - \lambda \Sigma^{-1}) \\
 &\quad \times (2\pi i)^{-1} \int_{c-i\infty}^{c+i\infty} v^{-s} \frac{\prod_{i=1}^m \Gamma\left[\frac{1}{2}n+s+k_i-\frac{1}{2}(i-1)\right]}{\prod_{i=1}^m \Gamma\left[\frac{1}{2}n+(k+i-1)m^{-1}+s\right]} ds. \quad (4.1)
 \end{aligned}$$

Recall that the G-function is defined as

$$\begin{aligned}
 G_{p,q}^{m,n} \left(v \left| \begin{array}{l} a_i, i=1, \dots, p \\ b_i, i=1, \dots, q \end{array} \right. \right) \\
 = (2\pi i)^{-1} \int_{c-i\infty}^{c+i\infty} \frac{\prod_{i=1}^m \Gamma(b_i-s) \prod_{i=1}^n \Gamma(1-a_i+s)}{\prod_{i=m+1}^q \Gamma(1-b_i+s) \prod_{i=n+1}^p \Gamma(a_i-s)} v^s ds;
 \end{aligned}$$

see Erdélyi et al. (1953). Then, by using the G-function, we have derived the following corollary.

COROLLARY 4.1. *The exact density of the likelihood ratio statistic V under the Gaussian model is given by*

$$\begin{aligned}
 f(v) &= \frac{\pi^{\frac{1}{4}m(m-1)} |\lambda \Sigma^{-1}|^{\frac{1}{2}n} (2\pi)^{\frac{1}{2}(m-1)}}{\Gamma_m\left(\frac{1}{2}n\right)} v^{\frac{1}{2}(n-m-1)} \\
 &\quad \times \sum_{k=0}^{\infty} \sum_{\kappa} \frac{1}{k!} m^{\frac{1}{2}-\frac{1}{2}mn-k} \Gamma\left(\frac{1}{2}mn+k\right) C_{\kappa}(\mathbf{I} - \lambda \Sigma^{-1}) G_{m,m}^{m,0} \left(v \left| \begin{array}{l} a_i \\ b_i \end{array} \right. \right),
 \end{aligned}$$

where $a_i = (k+i-1)m^{-1}+1/2n$ and $b_i = k_i+1/2(n-i+1)$ for $i = 1, \dots, m$.

A particular case of the above density, when $n = m - 1$ (a strong restriction), was first derived by Khatri and Srivastava (1971), based on their equation (2.9), p. 203, which is incorrect, but the subsequent equation (2.10) is right (compare with (4.1)) and the final particular density is also correct, see Khatri and Srivastava (1971, eq. (2.12)). They restricted the evaluation

of the G function for the particular $n = m - 1$. The computation of this particular exact density has not been discussed in the literature even though several works have appeared on the corresponding asymptotic distribution; see, for example, Muirhead (1982), and the references therein. Certainly, the above exact density of V appears to be not computable, but we see that it belongs to the following general expression $\sum_{k=0}^{\infty} \sum_{\kappa} g(k, \kappa) C_{\kappa}(\mathbf{X})$ which can be computed efficiently by a modification of the algorithms of Koev and Edelman (2006), and in fact at the same computational cost of the hypergeometric functions of one matrix argument. However, some important theoretical aspects must be solved first; we are working on this topic and we will report this research in a future work.

We can also analogously obtain the exact density of V associated with the non-Gaussian model considered earlier in Corollary 3.4.

COROLLARY 4.2. *The exact density of the likelihood ratio statistic V under the non-Gaussian model with generator function $h(y) = 2(mn)^{-1} \pi^{-mn/2} R^{1+mn/2} y e^{-Ry}$, $y > 0$, is given by*

$$\begin{aligned}
 f(v) = & \frac{\pi^{\frac{1}{4}(m-1)(m+2)} |\lambda \Sigma^{-1}|^{\frac{n}{2}}}{vn 2^{-\frac{1}{2}(m+1)} \Gamma_m(\frac{1}{2}n)} \sum_{k=0}^{\infty} \frac{\Gamma(k + \frac{mn}{2} + 1)}{k! m^{\frac{1}{2} + k + \frac{mn}{2}}} \\
 & \times \sum_{\kappa} C_{\kappa}(\mathbf{I} - \lambda \Sigma^{-1}) G_{2m, 2m}^{2m, 0} \left(v \left| \begin{array}{c} a_{1,i}, a_{2,i} \\ a_{3,i}, a_{4,i} \end{array} \right. \right) \\
 & + \frac{\pi^{\frac{1}{4}(m-1)(m+2)} |\lambda \Sigma^{-1}|^{\frac{n}{2}}}{vn 2^{-\frac{1}{2}(m+1)} \Gamma_m(\frac{1}{2}n)} \sum_{k=0}^{\infty} \frac{\Gamma(k + \frac{mn}{2} + 2)}{k! m^{\frac{5}{2} + k + \frac{mn}{2}}} \\
 & \times \sum_{\kappa} C_{\kappa}(\mathbf{I} - \lambda \Sigma^{-1}) G_{m, m}^{m, 0} \left(v \left| \begin{array}{c} b_{1,i} \\ b_{2,i} \end{array} \right. \right) \\
 & - \frac{\pi^{\frac{1}{4}(m-1)(m+2)} |\lambda \Sigma^{-1}|^{\frac{n}{2}}}{vn 2^{-\frac{1}{2}(m+1)} \Gamma_m(\frac{1}{2}n)} \sum_{k=0}^{\infty} \frac{\Gamma(k + \frac{mn}{2} + 1)}{k! m^{\frac{3}{2} + k + \frac{mn}{2}}} \\
 & \times \sum_{\kappa} C_{\kappa}(\mathbf{I} - \lambda \Sigma^{-1}) G_{m, m}^{m, 0} \left(v \left| \begin{array}{c} c_{1,i} \\ c_{2,i} \end{array} \right. \right),
 \end{aligned}$$

where $a_{1,i} = n/2 + (i - 1)m^{-1}$, $a_{2,i} = n/2 + (k + i)m^{-1}$, $a_{3,i} = 1/2n + k_i - 1/2(i - 1)$, $a_{4,i} = n/2 + im^{-1}$, $b_{1,i} = n/2 + (i + k + 1)m^{-1}$, $b_{2,i} = 1/2n + k_i - 1/2(i - 1)$, $c_{1,i} = n/2 + (k + i)m^{-1}$, $c_{2,i} = 1/2n + k_i - 1/2(i - 1)$ and $i = 1, \dots, m$.

PROOF. Start with the application of the correct version of the addressed equation (2.9) of Khatri and Srivastava (1971, p. 203)

$$\Gamma(ma) = (2\pi)^{-\frac{1}{2}(m-1)} m^{-\frac{1}{2}+ma} \prod_{i=1}^m \Gamma[a + (i-1)m^{-1}]$$

in Corollary 3.4; we obtain, after some algebraic simplification, that

$$f(v) = A + B - C,$$

where A, B and C are given by

$$\begin{aligned} A &= \frac{\pi^{\frac{1}{4}(m-1)(m+2)} |\lambda \Sigma^{-1}|^{\frac{n}{2}}}{vn 2^{-\frac{1}{2}(m+1)} \Gamma_m(\frac{1}{2}n)} \sum_{k=0}^{\infty} \frac{\Gamma(k + \frac{mn}{2} + 1)}{k! m^{\frac{1}{2}+k+\frac{mn}{2}}} \sum_{\kappa} C_{\kappa}(\mathbf{I} - \lambda \Sigma^{-1}) \\ &\quad \times (2\pi i)^{-1} \int_{c-i\infty}^{c+i\infty} v^s \frac{\prod_{i=1}^m \Gamma[\frac{1}{2}n - s + k_i - \frac{1}{2}(i-1)] \prod_{i=1}^m \Gamma[\frac{n}{2} - s + im^{-1}]}{\prod_{i=1}^m \Gamma[\frac{n}{2} - s + (i-1)m^{-1}] \prod_{i=1}^m \Gamma[\frac{n}{2} - s + (k+i)m^{-1}]} ds, \\ B &= \frac{\pi^{\frac{1}{4}(m-1)(m+2)} |\lambda \Sigma^{-1}|^{\frac{n}{2}}}{vn 2^{-\frac{1}{2}(m+1)} \Gamma_m(\frac{1}{2}n)} \sum_{k=0}^{\infty} \frac{\Gamma(k + \frac{mn}{2} + 2)}{k! m^{\frac{5}{2}+k+\frac{mn}{2}}} \sum_{\kappa} C_{\kappa}(\mathbf{I} - \lambda \Sigma^{-1}) \\ &\quad \times (2\pi i)^{-1} \int_{c-i\infty}^{c+i\infty} v^s \frac{\prod_{i=1}^m \Gamma[\frac{1}{2}n - s + k_i - \frac{1}{2}(i-1)]}{\prod_{i=1}^m \Gamma[\frac{n}{2} - s + (i+k+1)m^{-1}]} ds, \\ C &= \frac{\pi^{\frac{1}{4}(m-1)(m+2)} |\lambda \Sigma^{-1}|^{\frac{n}{2}}}{vn 2^{-\frac{1}{2}(m+1)} \Gamma_m(\frac{1}{2}n)} \sum_{k=0}^{\infty} \frac{\Gamma(k + \frac{mn}{2} + 1)}{k! m^{\frac{3}{2}+k+\frac{mn}{2}}} \sum_{\kappa} C_{\kappa}(\mathbf{I} - \lambda \Sigma^{-1}) \\ &\quad \times (2\pi i)^{-1} \int_{c-i\infty}^{c+i\infty} v^s \frac{\prod_{i=1}^m \Gamma[\frac{1}{2}n - s + k_i - \frac{1}{2}(i-1)]}{\prod_{i=1}^m \Gamma[\frac{n}{2} - s + (k+i)m^{-1}]} ds \end{aligned}$$

Upon using the definition of the G-function we obtain the required result.

Here again, we obtain a density proportional to $\sum_{k=0}^{\infty} \sum_{\kappa} g(k, \kappa) C_{\kappa}(\mathbf{X})$ which can be evaluated efficiently by a modification of the algorithm for hypergeometric functions due to Koev and Edelman (2006). However, this topic demands some new analytical developments which are currently under research.

In the literature, the sphericity test has been studied only for the Gaussian model; see Muirhead (1982), for example. Comparing the exact distributions of the test statistic V in Corollaries 4.1 and 4.2, we note that the densities are very different. In fact, this difference was noticed by Muirhead (1982) in the asymptotic case for a special set of models, in which the elliptical densities depend only on the kurtosis parameter.

Finally, even though the distribution function of V is exact and the series of zonal polynomials with partitionial coefficients can be computed, a natural problem that becomes of interest here is the computation of the G-function. This problem is currently under investigation for computing or simulating the exact distributions indexed by G-functions. However, the other densities or moments derived here that are not in terms of G-functions can be computed efficiently.

For example, Figure 1 presents the exact first three moments of V in the classical Gaussian case obtained from Corollary 3.1 under $W_3(4, \text{diag}(0.25, x, 1))$ and x running from 0.25 to 1. These plots are clearly different from the first three moments of V under the class of Kotz-type distributions indexed by $R > 0$ in Corollary 3.4 as shown in Figure 2 based on the same parameters under this model, i.e., $\mathbf{A}$ is $EW_3(4, \text{diag}(0.25, x, 1), h)$ and x running from 0.25 to 1. Figure 3 shows the exact pdf of $\text{tr } \mathbf{A}$, for $\mathbf{A} \sim W_3(4, \text{diag}(0.25, 0.5, 1))$, $\mathbf{A} \sim W_3(4, \text{diag}(1, 1, 1))$ and $\mathbf{A} \sim EW_3(4, \text{diag}(0.25, 0.5, 1), h)$, where h is the generator function of the Kotz-type model with parameters $R > 0, T = 0.1, 0.25, 0.5, 1, 2$. Finally, Figure 4 presents

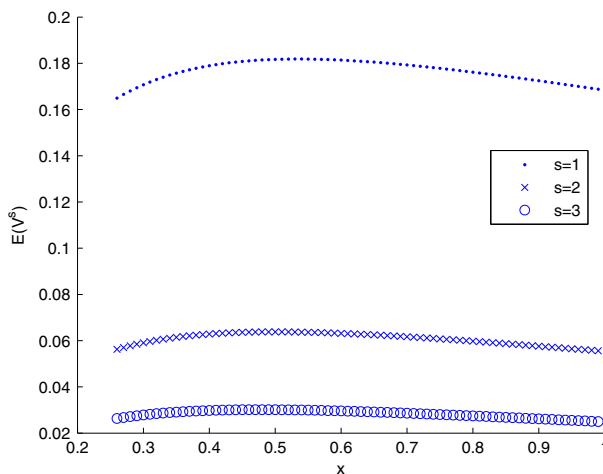


Figure 1: First moments of V under $\mathbf{A} \sim W_3(4, \text{diag}(0.25, x, 1))$, $0.25 < x < 1$.

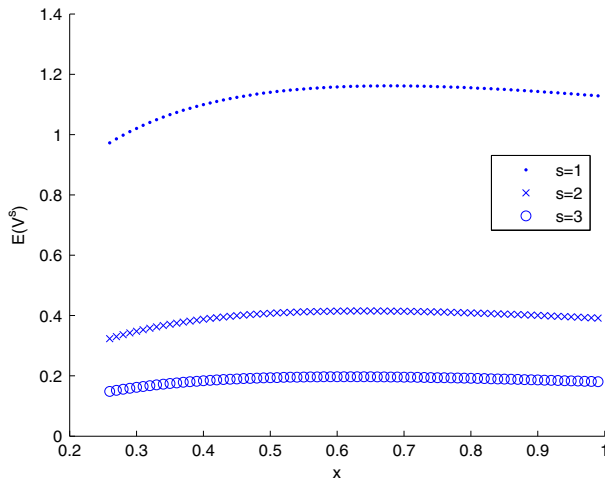


Figure 2: First moments of V under $\mathbf{A} \sim EW_3(4, \text{diag}(0.25, x, 1), \text{Kotz}(R > 0, T = 2))$, $0.25 < x < 1$.

the exact M.F.G $\phi(t)$, $t = -0.4$ of a generalized Wishart $\mathbf{A}$ under four matrices, $\mathbf{A} \sim W_3(4, \text{diag}(0.25, x, 1))$, $\mathbf{A} \sim EW_3(4, \text{diag}(0.25, x, 1), h)$, where h is the generator function of the Kotz-type model with parameters $R = 0.5$,

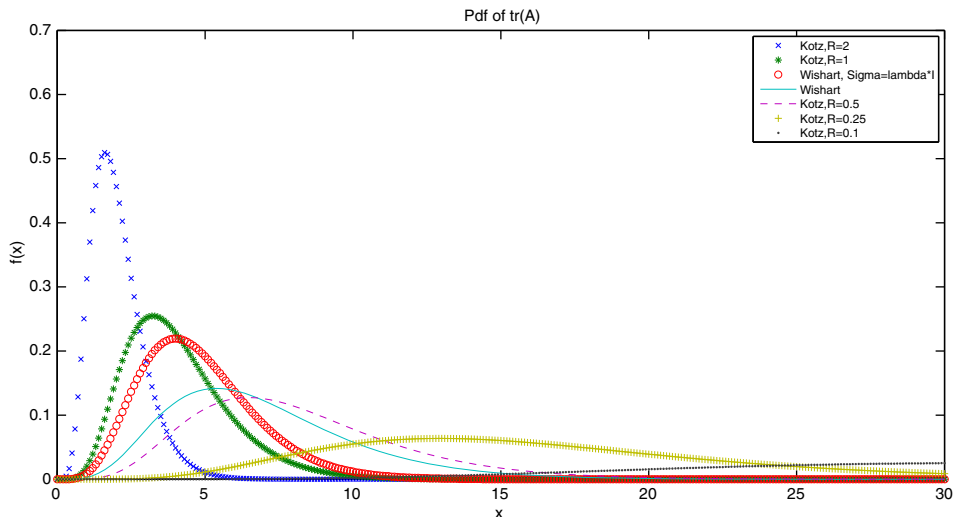


Figure 3: Pdf of $\text{tr } \mathbf{A}$, for $\mathbf{A} \sim W_3(4, \text{diag}(0.25, 0.5, 1))$, $\mathbf{A} \sim W_3(4, \text{diag}(1, 1, 1))$ and $\mathbf{A} \sim EW_3(4, \text{diag}(0.25, 0.5, 1), \text{Kotz}(R > 0, T = 0.1, 0.25, 0.5, 1, 2))$.

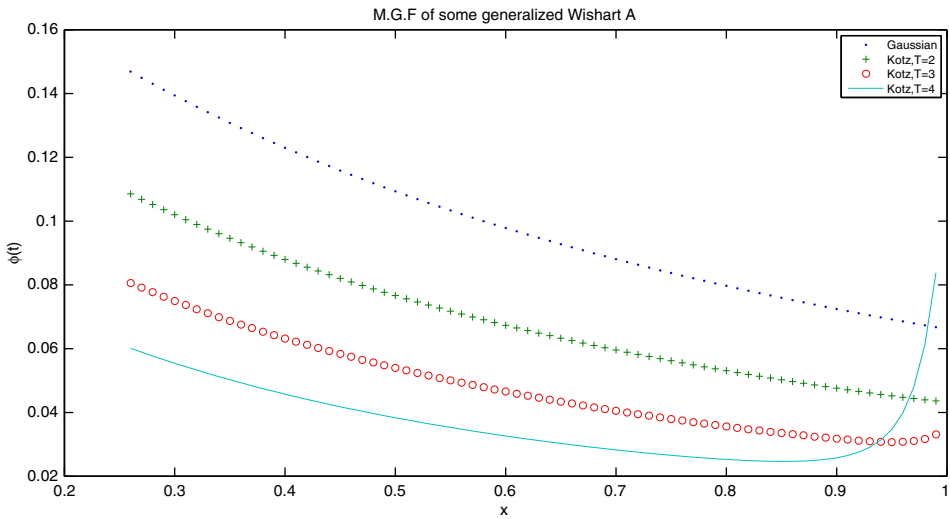


Figure 4: M.G.F. of $\mathbf{A}$, for $t = -0.4$, $0.25 < x < 1$, $\mathbf{A} \sim W_3(4, \text{diag}(0.25, x, 1))$, and $\mathbf{A} \sim EW_3(4, \text{diag}(0.25, x, 1), \text{Kotz}(R = 0.25, T = 2, 3, 4))$.

$T = 2, 3, 4$, and the variate x runs from 0.25 to 1. Again, the curves related with the generalized Wishart differ strongly from the classical Wishart distribution.

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